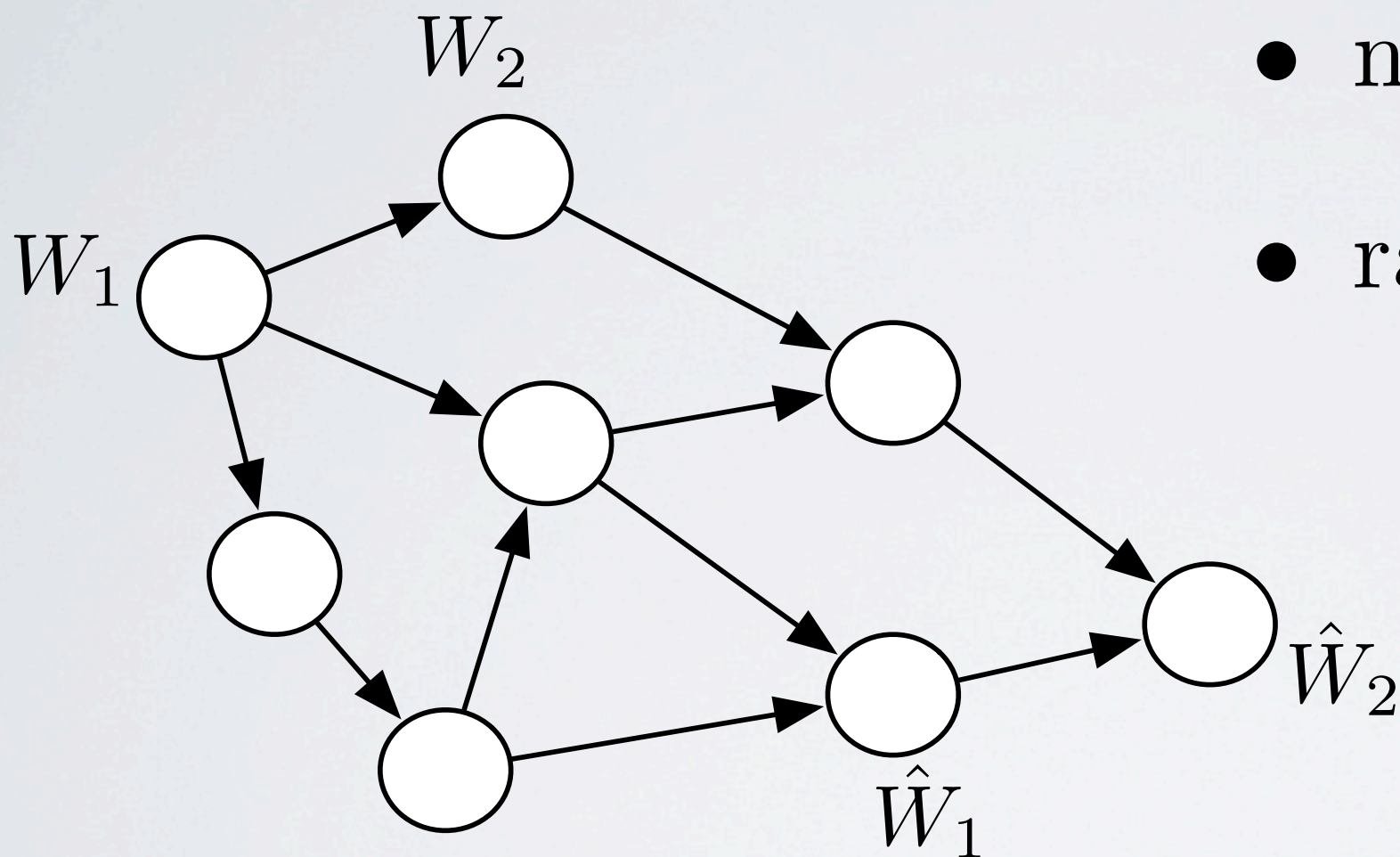


On Gaussian Interference Channels with Mixed Gaussian and Discrete inputs

Alex R Dytso, Daniela Tuninetti, Natasha Devroye,

Network Information Theory

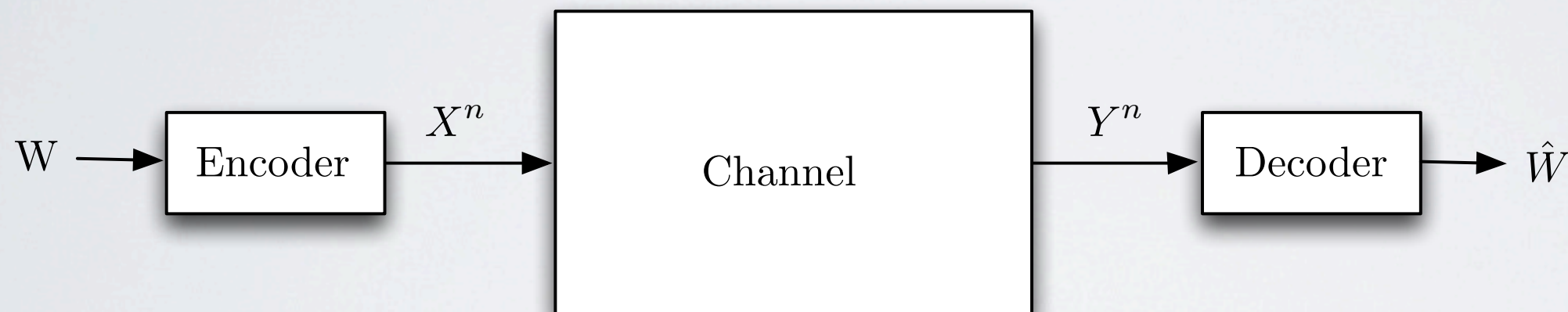


- n-transmission slots

- rate $R_i = \frac{\log |W_i|}{n}$

Point to Point Channel Example

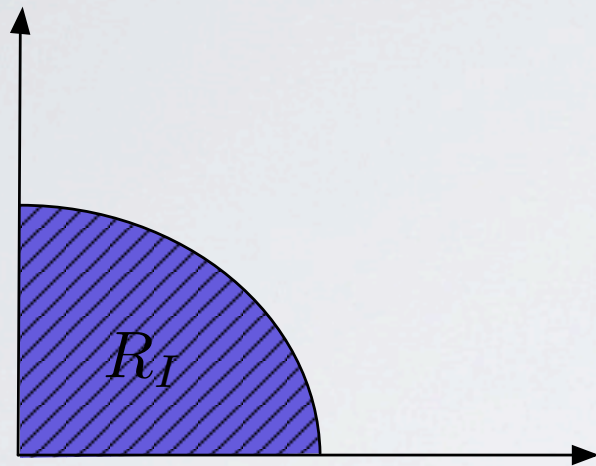
- n-transmission slots
- rate $R = \frac{\log |W|}{n}$



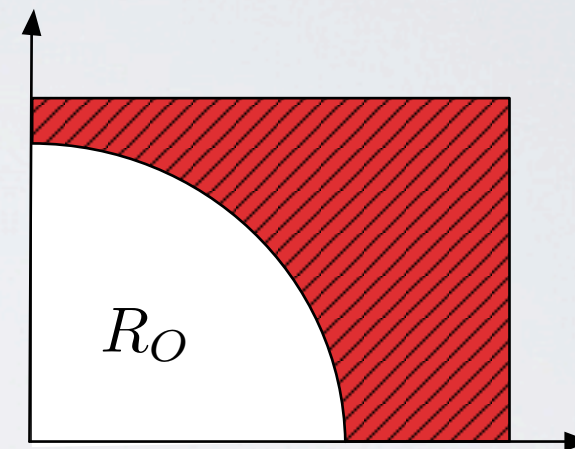
Capacity=the set of all achievable rates

Road map to channel capacity

Inner Bound / Achievability



Outer Bound



If $R_O = R_I$ we have capacity

Approximate Capacity

- In general capacity is difficult to find
- Fruitful direction is to approximate capacity

$$\text{gap} \geq R_O - R_I$$

Independent of

- input power
- Channel Gains magnitude
- Channel Phase

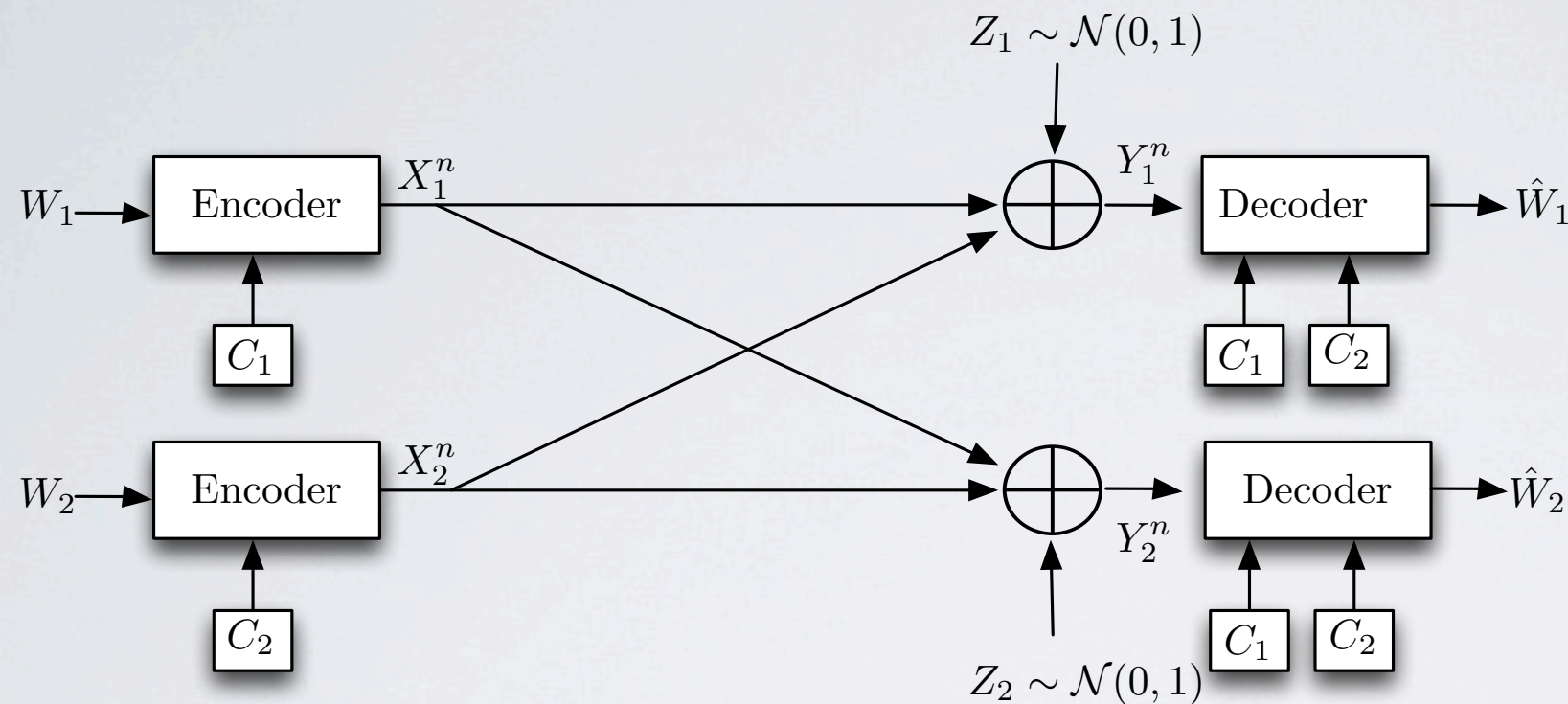
Mutual Information

$$I(X; Y) = \int p(x, y) \log\left(\frac{p(x, y)}{p(x)p(y)}\right) dx dy$$

For example capacity of point to point channel is

$$C = \max_{p(x)} I(X; Y)$$

Model Under Consideration



$$C_i = \begin{pmatrix} X_i^n(1) \\ X_i^n(2) \\ X_i^n(3) \\ \vdots \\ X_i^n(|W_i|) \end{pmatrix}$$

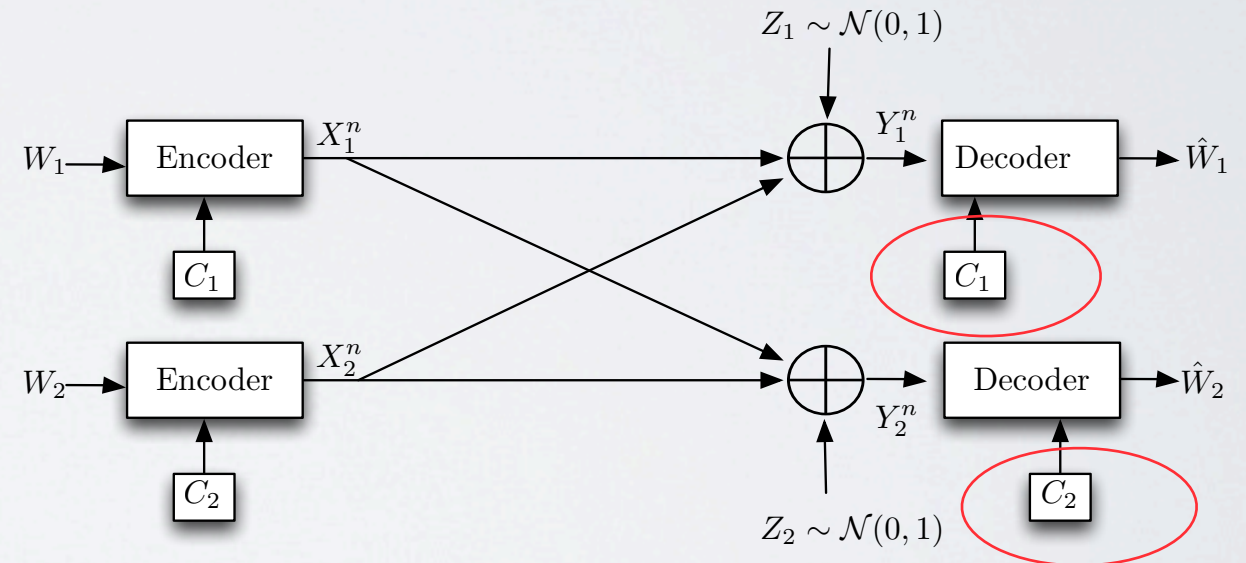
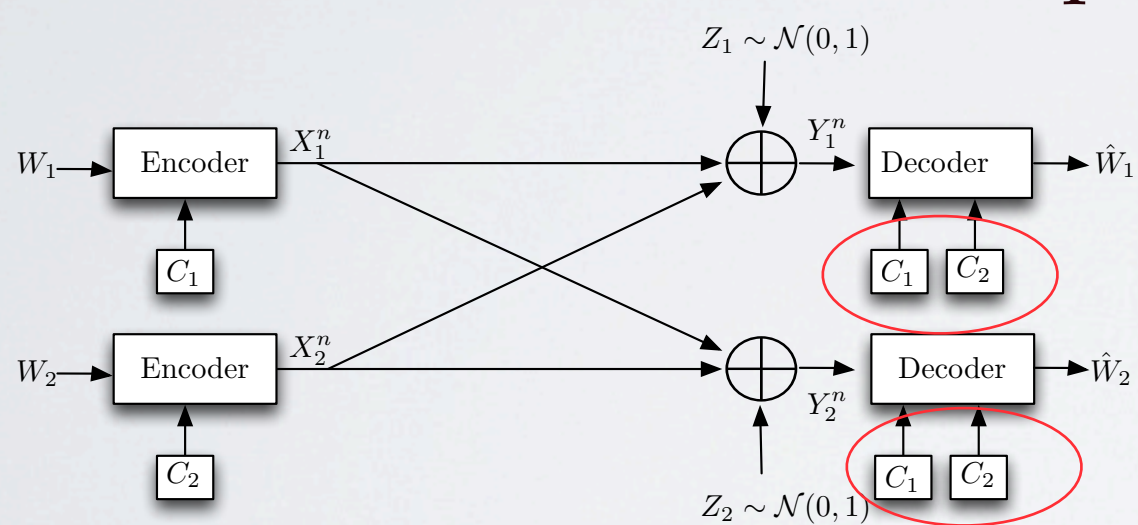
All Nodes are Synchronized

Solved to within gap=1bit

My Research Interests

Goal: To relax assumptions and make model more practical

- Relax Synchronism assumption
- Relax codebook assumption



Results so far

Theorem. Capacity of the asynchronous interference channel with lack of codebook knowledge is within $O(\log(\log(\text{SNR})))$ of capacity of the classical IC

SNR= strength of the direct link

$O(\log(\log(\text{SNR})))$ growth very slow

Surprising result: asynchronism and lack of codebook knowledge is not as harmful as one believes.

Key novelty

Use Mixed Gaussian Input

$$X_i = \sqrt{\delta} X_G + \sqrt{1 - \delta} X_D : \delta < 1$$

$$X_G \sim \mathcal{N}(0, 1), \quad X_D \sim \text{PAM}$$

Future Direction

- Extension to K-user cases
- Symbol asynchronism

Thank you